FA — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Nota sobre este algoritmo

Firefly Algorithm. CORREGIDO 2026-09-04: gastaba 39.6x el MaxFES y su atraccion hacia underflow a 0.0 (gamma sin reescalar). Estos resultados son los posteriores al arreglo.

Resumen — posicion entre los 10 algoritmos (rango por media)

dimension	presupuesto	rango medio de Friedman	posicion de 10	funciones donde es el mejor
D10	5k FES	8.62	9º	0 de 71
D10	50k FES	8.45	9º	0 de 71
D10	500k FES	6.21	6º	0 de 71
D50	5k FES	8.07	8º	0 de 71
D50	50k FES	8.08	9º	0 de 71
D50	500k FES	5.06	5º	0 de 71

FA — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	2.37e+06	3.87e+07	1.55e+07	1.57e+07	7.08e+06	4.36e+05	7.67e+06	3.58e+06	3.62e+06	1.66e+06	1.32e+03	9.81e+04	1.88e+04	2.52e+04	1.92e+04
F22014	2014	10	7.19e+08	2.11e+09	1.47e+09	1.45e+09	3.16e+08	1.61e+08	6.56e+08	4.83e+08	4.71e+08	1.16e+08	2.96e+04	7.98e+04	5.45e+04	5.38e+04	1.24e+04
F32014	2014	10	1.36e+04	5.80e+04	3.14e+04	3.08e+04	8.97e+03	5.33e+03	2.85e+04	1.23e+04	1.33e+04	5.60e+03	1.32e+00	7.68e+01	8.91e+00	1.14e+01	1.24e+01
F42014	2014	10	5.69e+01	1.30e+02	9.54e+01	9.27e+01	1.70e+01	3.89e+01	5.66e+01	4.74e+01	4.76e+01	4.35e+00	1.94e-02	3.48e+01	3.48e+01	2.10e+01	1.67e+01
F52014	2014	10	2.02e+01	2.08e+01	2.06e+01	2.06e+01	1.06e-01	1.84e+01	2.06e+01	2.04e+01	2.03e+01	3.24e-01	8.13e-01	2.04e+01	2.02e+01	1.91e+01	4.60e+00
F62014	2014	10	6.01e+00	1.01e+01	8.58e+00	8.43e+00	8.35e-01	4.73e+00	7.00e+00	6.11e+00	6.01e+00	5.90e-01	2.93e-01	2.74e+00	3.69e-01	7.44e-01	6.34e-01
F72014	2014	10	8.26e+00	2.52e+01	1.67e+01	1.70e+01	3.57e+00	2.46e+00	9.08e+00	6.53e+00	6.59e+00	1.36e+00	1.24e-01	4.40e-01	3.15e-01	3.05e-01	6.41e-02
F82014	2014	10	3.52e+01	6.54e+01	5.27e+01	5.17e+01	6.11e+00	2.09e+01	4.51e+01	3.61e+01	3.54e+01	5.05e+00	2.01e+00	1.69e+01	8.97e+00	9.02e+00	3.39e+00
F92014	2014	10	2.90e+01	6.81e+01	5.59e+01	5.44e+01	7.76e+00	2.75e+01	4.66e+01	3.88e+01	3.79e+01	4.90e+00	2.02e+00	1.89e+01	8.00e+00	8.29e+00	3.41e+00
F102014	2014	10	1.10e+03	1.89e+03	1.56e+03	1.56e+03	1.71e+02	7.34e+02	1.42e+03	1.11e+03	1.11e+03	1.52e+02	7.79e+00	3.92e+02	1.49e+02	1.52e+02	1.04e+02
F112014	2014	10	1.28e+03	2.07e+03	1.69e+03	1.70e+03	1.70e+02	9.73e+02	1.55e+03	1.26e+03	1.28e+03	1.48e+02	4.97e+00	7.81e+02	1.69e+02	2.18e+02	1.91e+02
F122014	2014	10	9.01e-01	2.62e+00	1.91e+00	1.86e+00	4.01e-01	6.54e-01	1.68e+00	1.28e+00	1.22e+00	2.78e-01	6.11e-02	1.98e-01	9.36e-02	9.80e-02	2.49e-02
F132014	2014	10	6.51e-01	1.37e+00	1.06e+00	1.05e+00	1.44e-01	2.67e-01	7.36e-01	5.91e-01	5.72e-01	1.03e-01	2.98e-02	1.39e-01	7.93e-02	8.17e-02	2.29e-02
F142014	2014	10	1.54e+00	7.33e+00	4.08e+00	4.15e+00	1.22e+00	3.29e-01	1.49e+00	8.63e-01	8.53e-01	2.85e-01	1.50e-02	1.92e-01	8.03e-02	7.98e-02	3.84e-02
F152014	2014	10	1.19e+01	2.48e+02	6.48e+01	7.24e+01	4.87e+01	6.99e+00	1.72e+01	1.09e+01	1.12e+01	2.12e+00	1.09e-01	1.29e+00	8.05e-01	8.06e-01	1.96e-01
F162014	2014	10	3.09e+00	4.08e+00	3.82e+00	3.77e+00	1.96e-01	2.55e+00	3.66e+00	3.33e+00	3.30e+00	2.34e-01	3.78e-01	2.76e+00	1.65e+00	1.68e+00	5.94e-01
F172014	2014	10	4.07e+04	5.27e+05	1.78e+05	1.89e+05	1.08e+05	1.07e+04	1.31e+05	4.09e+04	4.84e+04	2.85e+04	9.05e+02	2.14e+04	4.95e+03	5.64e+03	4.31e+03
F182014	2014	10	2.46e+04	4.03e+06	4.03e+05	5.87e+05	6.78e+05	2.46e+04	1.77e+05	8.44e+04	9.03e+04	3.15e+04	8.74e+03	6.37e+04	2.76e+04	2.72e+04	1.20e+04
F192014	2014	10	2.51e+01	1.24e+04	7.57e+02	1.47e+03	2.09e+03	1.48e+01	2.96e+02	7.72e+01	8.64e+01	5.85e+01	3.09e+00	1.63e+01	7.80e+00	8.76e+00	3.38e+00
F202014	2014	10	2.18e+06	3.12e+08	5.80e+07	7.63e+07	7.36e+07	2.44e+04	6.63e+06	1.26e+06	1.51e+06	1.55e+06	1.41e+03	3.60e+04	1.27e+04	1.43e+04	8.83e+03
F212014	2014	10	3.22e+04	1.52e+06	3.26e+05	4.43e+05	3.43e+05	1.64e+04	2.45e+05	6.21e+04	7.51e+04	4.67e+04	4.00e+02	2.54e+04	6.01e+03	6.60e+03	4.99e+03
F222014	2014	10	4.18e+02	3.44e+03	9.89e+02	1.09e+03	5.56e+02	9.17e+01	8.55e+02	4.00e+02	3.91e+02	1.57e+02	2.26e+01	2.42e+02	2.37e+01	7.88e+01	9.55e+01
F232014	2014	10	2.60e+02	3.00e+02	2.79e+02	2.80e+02	9.75e+00	1.89e+02	2.61e+02	2.56e+02	2.53e+02	1.11e+01	7.22e-01	2.42e+02	2.42e+02	2.27e+02	5.73e+01
F242014	2014	10	1.26e+02	1.76e+02	1.68e+02	1.65e+02	8.30e+00	1.26e+02	1.55e+02	1.47e+02	1.48e+02	5.28e+00	1.07e+02	1.22e+02	1.14e+02	1.15e+02	3.78e+00
F252014	2014	10	1.98e+02	2.07e+02	2.04e+02	2.04e+02	1.87e+00	1.71e+02	2.02e+02	2.01e+02	1.98e+02	9.32e+00	1.34e+02	2.00e+02	2.00e+02	1.93e+02	2.03e+01
F262014	2014	10	1.01e+02	1.01e+02	1.01e+02	1.01e+02	1.64e-01	1.00e+02	1.01e+02	1.01e+02	1.01e+02	8.28e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	2.53e-02
F272014	2014	10	5.33e+01	5.56e+02	4.20e+02	4.00e+02	1.27e+02	1.20e+01	4.80e+02	4.05e+02	3.64e+02	1.40e+02	6.93e-01	4.00e+02	4.00e+02	3.26e+02	1.36e+02
F282014	2014	10	4.12e+02	7.10e+02	4.38e+02	4.83e+02	8.05e+01	3.80e+02	5.65e+02	4.06e+02	4.40e+02	5.66e+01	3.57e+02	5.03e+02	3.69e+02	4.05e+02	5.51e+01
F292014	2014	10	5.25e+06	1.92e+07	1.07e+07	1.13e+07	3.10e+06	5.17e+06	1.37e+07	6.14e+06	6.71e+06	1.53e+06	4.50e+06	1.29e+07	5.31e+06	5.81e+06	1.40e+06
F302014	2014	10	2.01e+05	1.02e+07	1.80e+06	2.55e+06	2.27e+06	1.89e+04	1.10e+06	1.31e+05	1.92e+05	1.83e+05	2.34e+03	2.16e+05	1.33e+04	3.35e+04	3.90e+04
F12017	2017	10	3.85e+08	2.42e+09	1.31e+09	1.28e+09	4.02e+08	1.88e+08	6.16e+08	4.34e+08	4.38e+08	9.49e+07	2.24e+04	8.98e+04	5.51e+04	5.41e+04	1.42e+04
F22017	2017	10	1.61e+03	1.04e+04	4.32e+03	4.47e+03	1.76e+03	3.87e+02	1.73e+03	9.95e+02	1.02e+03	2.95e+02	4.99e-02	1.57e-01	1.02e-01	9.85e-02	2.31e-02
F32017	2017	10	4.14e+01	1.96e+02	9.33e+01	9.85e+01	2.69e+01	2.53e+01	5.02e+01	3.95e+01	3.85e+01	6.36e+00	1.55e-01	4.56e+00	4.29e+00	2.82e+00	1.98e+00
F42017	2017	10	5.03e+02	2.74e+03	1.93e+03	1.91e+03	5.14e+02	4.57e+02	9.56e+02	7.02e+02	7.08e+02	1.29e+02	9.26e+00	3.55e+01	2.00e+01	2.01e+01	4.37e+00
F52017	2017	10	3.35e-04	4.14e-03	1.34e-03	1.73e-03	1.06e-03	7.83e-06	6.31e-04	2.32e-04	2.74e-04	1.51e-04	4.54e-11	1.41e-07	5.08e-10	4.74e-09	2.13e-08
F62017	2017	10	1.97e+04	6.87e+04	3.91e+04	3.99e+04	1.01e+04	5.43e+03	2.23e+04	1.39e+04	1.36e+04	3.68e+03	3.87e+01	5.81e+01	5.10e+01	4.94e+01	5.56e+00
F72017	2017	10	4.25e+01	1.17e+02	8.20e+01	8.15e+01	1.83e+01	2.68e+01	6.40e+01	4.36e+01	4.32e+01	8.60e+00	1.00e+01	2.80e+01	1.60e+01	1.72e+01	4.47e+00
F82017	2017	10	8.74e-01	2.25e+00	1.63e+00	1.63e+00	3.42e-01	3.37e-01	7.37e-01	5.33e-01	5.28e-01	9.92e-02	6.09e-05	1.46e-04	1.02e-04	1.00e-04	2.20e-05
F92017	2017	10	9.57e+02	1.95e+03	1.66e+03	1.61e+03	2.19e+02	7.48e+02	1.60e+03	1.22e+03	1.22e+03	1.76e+02	7.58e+00	4.74e+02	1.56e+02	1.65e+02	1.32e+02
F102017	2017	10	2.63e+04	1.07e+06	2.00e+05	2.22e+05	1.79e+05	8.00e+03	8.20e+04	3.63e+04	3.93e+04	1.65e+04	5.78e+00	1.56e+04	1.12e+01	4.89e+02	2.21e+03
F112017	2017	10	5.40e+07	2.26e+08	1.10e+08	1.10e+08	3.96e+07	9.16e+06	6.16e+07	2.45e+07	2.76e+07	1.19e+07	6.09e+03	6.64e+04	2.53e+04	2.99e+04	1.73e+04
F122017	2017	10	1.92e+05	1.35e+07	5.33e+06	6.00e+06	3.54e+06	1.23e+05	1.88e+06	6.90e+05	6.92e+05	3.97e+05	1.94e+03	2.34e+04	9.14e+03	9.39e+03	4.80e+03
F132017	2017	10	7.02e+03	4.01e+04	1.73e+04	1.84e+04	8.05e+03	1.11e+03	1.75e+04	6.02e+03	7.34e+03	4.38e+03	3.89e+01	2.75e+03	1.48e+02	3.61e+02	4.91e+02
F142017	2017	10	2.51e+04	1.86e+06	4.34e+05	4.93e+05	3.73e+05	2.32e+04	3.13e+05	5.70e+04	7.40e+04	5.79e+04	5.40e+02	1.85e+04	5.88e+03	6.50e+03	4.49e+03
F152017	2017	10	3.16e+03	4.78e+04	1.15e+04	1.48e+04	9.94e+03	1.47e+03	1.12e+04	4.20e+03	4.63e+03	2.14e+03	1.45e+00	9.20e+02	4.98e+00	1.79e+02	2.20e+02
F162017	2017	10	1.08e+03	7.20e+04	8.22e+03	1.40e+04	1.43e+04	4.85e+02	9.00e+03	1.53e+03	1.96e+03	1.44e+03	1.49e+01	6.52e+02	3.04e+01	1.09e+02	1.46e+02
F172017	2017	10	4.30e+04	1.61e+06	2.41e+05	4.10e+05	3.91e+05	9.12e+03	1.64e+05	5.87e+04	6.30e+04	3.31e+04	2.48e+03	3.57e+04	1.29e+04	1.47e+04	9.04e+03
F182017	2017	10	5.98e+04	6.10e+07	6.07e+06	1.15e+07	1.42e+07	2.00e+03	6.10e+05	9.18e+04	1.40e+05	1.41e+05	9.04e+00	5.34e+03	7.12e+02	1.33e+03	1.44e+03
F192017	2017	10	2.07e+02	7.28e+02	4.81e+02	4.68e+02	1.22e+02	8.04e+01	2.81e+02	1.84e+02	1.85e+02	5.01e+01	8.70e+00	8.96e+01	2.84e+01	3.62e+01	1.72e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	5.76e+02	1.78e+03	6.80e+02	7.02e+02	1.62e+02	3.43e+02	7.77e+02	4.67e+02	4.68e+02	6.75e+01	1.03e+02	2.11e+02	1.04e+02	1.06e+02	1.50e+01
F212017	2017	10	1.17e+02	1.23e+02	1.21e+02	1.21e+02	1.45e+00	1.14e+02	1.17e+02	1.16e+02	1.16e+02	8.00e-01	1.00e+02	1.04e+02	1.00e+02	1.01e+02	1.10e+00
F222017	2017	10	7.42e+02	1.25e+03	9.86e+02	9.86e+02	1.09e+02	5.11e+02	7.92e+02	6.76e+02	6.67e+02	5.77e+01	1.07e+02	2.05e+02	1.12e+02	1.20e+02	2.82e+01
F232017	2017	10	1.51e+03	2.34e+03	1.91e+03	1.91e+03	1.63e+02	8.69e+02	1.43e+03	1.23e+03	1.22e+03	1.45e+02	1.12e+02	3.23e+02	2.12e+02	2.12e+02	2.95e+01
F242017	2017	10	5.21e+02	6.33e+02	5.71e+02	5.77e+02	2.49e+01	4.94e+02	5.30e+02	5.12e+02	5.12e+02	8.41e+00	4.77e+02	4.80e+02	4.80e+02	4.79e+02	1.00e+00
F252017	2017	10	1.50e+02	5.57e+02	4.78e+02	4.72e+02	4.81e+01	5.11e+01	4.73e+02	4.70e+02	4.36e+02	1.10e+02	2.99e-01	4.68e+02	4.67e+02	4.28e+02	1.26e+02
F262017	2017	10	4.01e+02	4.11e+02	4.07e+02	4.07e+02	2.21e+00	3.97e+02	4.04e+02	4.01e+02	4.01e+02	1.58e+00	3.90e+02	3.96e+02	3.91e+02	3.91e+02	1.15e+00
F272017	2017	10	2.42e+02	4.91e+02	3.03e+02	3.06e+02	3.87e+01	1.79e+02	2.36e+02	2.17e+02	2.15e+02	1.28e+01	3.90e+00	9.59e+00	6.97e+00	6.99e+00	1.44e+00
F282017	2017	10	6.44e+03	8.10e+05	7.58e+04	1.39e+05	1.60e+05	2.01e+03	3.87e+04	9.89e+03	1.22e+04	9.91e+03	9.86e+02	1.36e+03	1.00e+03	1.04e+03	9.08e+01
F292017	2017	10	2.16e+06	1.10e+08	3.19e+07	3.74e+07	2.45e+07	6.17e+05	9.16e+06	2.18e+06	2.98e+06	2.07e+06	4.46e+03	5.03e+04	1.52e+04	1.55e+04	7.92e+03
F12022	2022	10	1.29e+03	5.90e+03	3.21e+03	3.28e+03	1.04e+03	5.04e+02	1.55e+03	9.20e+02	9.83e+02	2.35e+02	3.03e-02	1.31e-01	8.05e-02	7.79e-02	2.11e-02
F22022	2022	10	4.79e+01	1.33e+02	7.23e+01	7.40e+01	1.84e+01	1.52e+01	4.08e+01	2.96e+01	2.91e+01	5.13e+00	4.08e+00	8.92e+00	8.92e+00	7.24e+00	2.30e+00
F32022	2022	10	2.50e-02	1.02e-01	7.36e-02	7.24e-02	1.58e-02	1.01e-02	3.68e-02	2.43e-02	2.52e-02	6.07e-03	1.43e-06	4.52e-06	2.91e-06	2.95e-06	7.53e-07
F42022	2022	10	4.37e+01	1.06e+02	7.59e+01	7.52e+01	1.67e+01	2.67e+01	6.01e+01	4.53e+01	4.40e+01	8.04e+00	1.20e+01	2.80e+01	1.60e+01	1.70e+01	3.50e+00
F52022	2022	10	3.09e-01	1.85e+00	1.06e+00	1.06e+00	3.31e-01	2.66e-01	5.93e-01	3.94e-01	4.00e-01	6.42e-02	3.40e-05	1.13e-04	7.36e-05	7.61e-05	2.03e-05
F62022	2022	10	6.57e+05	2.72e+07	8.22e+06	9.74e+06	6.04e+06	1.80e+05	5.40e+06	1.32e+06	1.51e+06	1.02e+06	3.02e+03	9.33e+04	4.34e+04	4.40e+04	1.96e+04
F72022	2022	10	7.08e+01	3.27e+02	1.78e+02	1.88e+02	5.84e+01	5.43e+01	1.94e+02	8.85e+01	9.41e+01	2.76e+01	7.67e+00	1.27e+02	2.57e+01	2.62e+01	1.53e+01
F82022	2022	10	2.05e+02	3.96e+03	9.52e+02	1.18e+03	7.37e+02	6.11e+01	1.18e+03	2.90e+02	3.37e+02	2.29e+02	4.89e+00	1.23e+02	2.41e+01	2.46e+01	1.48e+01
F92022	2022	10	1.79e+02	4.72e+02	3.98e+02	3.95e+02	5.38e+01	6.80e+01	3.79e+02	1.26e+02	1.69e+02	1.05e+02	1.52e-01	3.59e+02	2.67e-01	6.36e+01	1.38e+02
F102022	2022	10	2.41e+02	4.31e+02	2.50e+02	2.66e+02	5.19e+01	2.29e+02	3.84e+02	2.37e+02	2.50e+02	4.20e+01	2.21e+02	3.25e+02	2.21e+02	2.31e+02	2.97e+01
F112022	2022	10	2.68e+01	9.01e+02	3.87e+01	1.06e+02	2.31e+02	1.62e+01	8.79e+02	2.36e+01	9.03e+01	2.32e+02	8.83e-02	8.76e+02	1.19e-01	6.88e+01	2.38e+02
F122022	2022	10	1.69e+02	1.76e+02	1.72e+02	1.72e+02	1.67e+00	1.68e+02	1.71e+02	1.69e+02	1.69e+02	6.08e-01	1.64e+02	1.66e+02	1.64e+02	1.65e+02	6.66e-01

FA — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	8.51e+08	2.02e+09	1.42e+09	1.43e+09	2.60e+08	3.35e+08	7.82e+08	4.88e+08	5.04e+08	9.64e+07	3.26e+06	1.16e+07	5.45e+06	5.95e+06	1.88e+06
F22014	2014	50	5.32e+10	8.60e+10	7.11e+10	7.13e+10	6.89e+09	2.35e+10	3.67e+10	2.99e+10	2.99e+10	2.74e+09	1.79e+06	5.34e+06	3.19e+06	3.20e+06	6.51e+05
F32014	2014	50	1.99e+05	3.43e+05	2.58e+05	2.64e+05	3.33e+04	1.00e+05	2.48e+05	1.80e+05	1.80e+05	2.89e+04	3.33e+03	2.01e+04	7.58e+03	8.45e+03	3.45e+03
F42014	2014	50	5.33e+03	1.62e+04	1.10e+04	1.11e+04	2.41e+03	1.63e+03	3.75e+03	2.43e+03	2.44e+03	4.17e+02	7.57e+01	1.94e+02	9.89e+01	9.89e+01	1.49e+01
F52014	2014	50	2.12e+01	2.13e+01	2.13e+01	2.13e+01	3.58e-02	2.11e+01	2.13e+01	2.12e+01	2.12e+01	3.72e-02	2.05e+01	2.12e+01	2.11e+01	2.11e+01	1.29e-01
F62014	2014	50	6.69e+01	7.60e+01	7.10e+01	7.11e+01	2.04e+00	5.58e+01	6.80e+01	6.02e+01	6.06e+01	2.29e+00	4.16e+00	2.13e+01	1.44e+01	1.40e+01	3.62e+00
F72014	2014	50	5.16e+02	9.55e+02	7.57e+02	7.49e+02	9.72e+01	2.29e+02	3.99e+02	2.92e+02	2.97e+02	3.62e+01	9.64e-01	1.06e+00	1.02e+00	1.02e+00	1.79e-02
F82014	2014	50	4.72e+02	6.61e+02	6.02e+02	5.94e+02	3.94e+01	4.28e+02	5.49e+02	4.70e+02	4.70e+02	2.23e+01	6.08e+01	1.63e+02	1.07e+02	1.08e+02	2.33e+01
F92014	2014	50	6.08e+02	7.84e+02	6.89e+02	6.92e+02	3.71e+01	4.62e+02	5.79e+02	5.35e+02	5.35e+02	2.19e+01	6.31e+01	1.68e+02	1.19e+02	1.17e+02	2.45e+01
F102014	2014	50	1.34e+04	1.52e+04	1.46e+04	1.45e+04	4.42e+02	1.25e+04	1.43e+04	1.37e+04	1.36e+04	4.17e+02	2.42e+03	6.30e+03	4.68e+03	4.53e+03	8.89e+02
F112014	2014	50	1.37e+04	1.57e+04	1.50e+04	1.49e+04	4.10e+02	1.33e+04	1.47e+04	1.40e+04	1.40e+04	3.86e+02	2.56e+03	5.86e+03	4.12e+03	4.13e+03	6.25e+02
F122014	2014	50	3.51e+00	5.88e+00	4.81e+00	4.79e+00	5.02e-01	2.99e+00	4.55e+00	4.00e+00	3.94e+00	3.82e-01	8.94e-02	7.58e-01	3.70e-01	3.96e-01	1.41e-01
F132014	2014	50	4.84e+00	6.22e+00	5.50e+00	5.51e+00	3.17e-01	2.13e+00	3.78e+00	3.35e+00	3.29e+00	2.78e-01	3.48e-01	6.80e-01	4.85e-01	4.86e-01	8.67e-02
F142014	2014	50	1.53e+02	2.67e+02	2.00e+02	1.98e+02	2.49e+01	6.34e+01	1.01e+02	7.91e+01	7.98e+01	7.12e+00	1.77e-01	1.04e+00	2.65e-01	4.65e-01	3.28e-01
F152014	2014	50	5.49e+05	6.87e+06	2.41e+06	2.63e+06	1.35e+06	1.64e+04	2.12e+05	5.17e+04	5.73e+04	3.21e+04	1.64e+01	2.91e+01	2.09e+01	2.12e+01	2.94e+00
F162014	2014	50	2.26e+01	2.37e+01	2.33e+01	2.33e+01	2.30e-01	2.24e+01	2.31e+01	2.28e+01	2.28e+01	1.83e-01	1.92e+01	2.24e+01	2.07e+01	2.06e+01	6.54e-01
F172014	2014	50	4.14e+07	1.71e+08	9.37e+07	9.36e+07	2.80e+07	1.28e+07	5.60e+07	3.19e+07	3.19e+07	8.71e+06	2.14e+05	1.22e+06	4.71e+05	5.14e+05	2.16e+05
F182014	2014	50	8.52e+08	4.20e+09	2.81e+09	2.78e+09	6.29e+08	3.87e+08	1.33e+09	9.99e+08	9.76e+08	1.98e+08	2.40e+05	4.30e+05	3.51e+05	3.49e+05	4.66e+04
F192014	2014	50	1.09e+08	7.79e+08	3.51e+08	3.31e+08	1.32e+08	8.12e+06	7.10e+07	3.44e+07	3.54e+07	1.64e+07	6.10e+01	1.18e+04	9.73e+01	9.54e+02	2.34e+03
F202014	2014	50	4.25e+11	4.51e+13	1.07e+13	1.29e+13	1.09e+13	2.08e+10	4.12e+11	1.65e+11	1.63e+11	8.82e+10	1.17e+05	3.97e+05	2.59e+05	2.52e+05	5.52e+04
F212014	2014	50	1.46e+08	8.42e+08	4.34e+08	4.28e+08	1.40e+08	4.21e+07	1.74e+08	1.04e+08	9.97e+07	2.74e+07	1.51e+05	1.72e+06	4.22e+05	4.99e+05	2.81e+05
F222014	2014	50	1.78e+10	1.80e+12	3.09e+11	3.81e+11	3.27e+11	6.42e+07	1.32e+10	2.34e+09	3.00e+09	2.60e+09	2.70e+02	2.12e+03	7.90e+02	8.32e+02	3.44e+02
F232014	2014	50	5.36e+02	9.64e+02	7.31e+02	7.27e+02	8.81e+01	3.87e+02	4.91e+02	4.39e+02	4.40e+02	2.52e+01	2.35e+02	2.35e+02	2.35e+02	2.35e+02	6.54e-03
F242014	2014	50	5.39e+02	6.34e+02	5.99e+02	5.98e+02	2.23e+01	4.23e+02	4.84e+02	4.50e+02	4.50e+02	1.46e+01	2.58e+02	3.00e+02	2.90e+02	2.86e+02	1.15e+01
F252014	2014	50	3.32e+02	4.28e+02	3.71e+02	3.78e+02	2.20e+01	2.60e+02	3.19e+02	2.88e+02	2.86e+02	1.12e+01	2.07e+02	2.13e+02	2.09e+02	2.10e+02	1.48e+00
F262014	2014	50	1.05e+02	4.87e+02	2.92e+02	2.27e+02	1.20e+02	1.03e+02	4.55e+02	1.03e+02	1.52e+02	8.78e+01	1.00e+02	3.43e+02	1.01e+02	1.33e+02	5.96e+01
F272014	2014	50	2.04e+03	2.28e+03	2.16e+03	2.15e+03	5.11e+01	1.74e+03	2.00e+03	1.86e+03	1.86e+03	5.71e+01	5.48e+02	8.85e+02	7.19e+02	7.09e+02	7.73e+01
F282014	2014	50	3.70e+03	8.05e+03	6.51e+03	6.10e+03	1.23e+03	2.17e+03	5.07e+03	2.62e+03	3.13e+03	9.34e+02	1.14e+03	2.10e+03	1.29e+03	1.35e+03	1.96e+02
F292014	2014	50	9.64e+08	2.92e+09	1.77e+09	1.75e+09	4.33e+08	2.18e+08	5.97e+08	3.64e+08	3.70e+08	9.20e+07	1.15e+04	2.37e+05	2.41e+04	3.48e+04	4.14e+04
F302014	2014	50	6.40e+09	7.72e+11	2.11e+11	2.46e+11	1.77e+11	9.32e+08	2.73e+10	7.11e+09	8.28e+09	5.12e+09	1.64e+05	4.15e+06	4.25e+05	5.94e+05	6.19e+05
F12017	2017	50	4.86e+10	8.16e+10	6.91e+10	6.89e+10	7.15e+09	1.83e+10	3.53e+10	2.88e+10	2.85e+10	3.49e+09	2.10e+06	4.31e+06	3.13e+06	3.04e+06	5.57e+05
F22017	2017	50	1.31e+05	2.26e+05	1.77e+05	1.76e+05	2.38e+04	4.43e+04	7.85e+04	6.30e+04	6.16e+04	7.99e+03	4.50e+00	8.74e+00	6.49e+00	6.64e+00	9.91e-01
F32017	2017	50	6.45e+03	1.78e+04	1.08e+04	1.12e+04	2.57e+03	1.44e+03	3.29e+03	2.40e+03	2.32e+03	4.09e+02	5.64e+01	1.69e+02	1.34e+02	1.10e+02	4.27e+01
F42017	2017	50	6.03e+04	1.09e+05	7.53e+04	7.74e+04	9.70e+03	2.09e+04	3.61e+04	2.85e+04	2.86e+04	3.18e+03	3.37e+02	4.58e+02	3.85e+02	3.86e+02	2.73e+01
F52017	2017	50	2.52e-02	7.03e-02	5.47e-02	5.44e-02	9.20e-03	1.67e-02	4.42e-02	2.76e-02	2.83e-02	6.18e-03	2.49e-07	1.05e-04	5.81e-06	1.81e-05	2.52e-05
F62017	2017	50	1.80e+06	3.18e+06	2.56e+06	2.57e+06	3.29e+05	8.07e+05	1.19e+06	1.01e+06	1.01e+06	8.45e+04	5.92e+02	7.61e+02	6.75e+02	6.69e+02	4.20e+01
F72017	2017	50	1.08e+03	1.49e+03	1.33e+03	1.33e+03	8.74e+01	8.92e+02	1.17e+03	1.02e+03	1.01e+03	6.32e+01	2.08e+02	5.40e+02	3.00e+02	3.06e+02	6.93e+01
F82017	2017	50	5.01e+01	1.09e+02	7.75e+01	7.88e+01	1.28e+01	1.55e+01	3.76e+01	2.78e+01	2.76e+01	5.27e+00	3.59e-03	5.10e+00	1.37e+00	1.69e+00	1.40e+00
F92017	2017	50	1.40e+04	1.57e+04	1.51e+04	1.50e+04	4.95e+02	1.31e+04	1.48e+04	1.42e+04	1.42e+04	4.01e+02	2.77e+03	5.85e+03	4.50e+03	4.50e+03	6.52e+02
F102017	2017	50	6.37e+08	2.44e+09	1.40e+09	1.45e+09	4.42e+08	1.03e+08	3.90e+08	2.12e+08	2.13e+08	6.55e+07	2.35e+04	2.97e+05	1.34e+05	1.26e+05	6.86e+04
F112017	2017	50	6.72e+09	1.16e+10	8.92e+09	8.94e+09	1.07e+09	2.22e+09	4.36e+09	3.42e+09	3.34e+09	5.36e+08	2.34e+06	3.81e+07	1.10e+07	1.34e+07	8.97e+06
F122017	2017	50	1.18e+10	2.88e+10	1.79e+10	1.81e+10	3.52e+09	3.22e+09	7.17e+09	5.48e+09	5.46e+09	9.97e+08	5.54e+05	1.62e+06	9.19e+05	9.69e+05	2.47e+05
F132017	2017	50	5.54e+06	4.14e+07	1.62e+07	1.82e+07	7.84e+06	1.37e+06	9.83e+06	5.94e+06	5.87e+06	1.88e+06	1.37e+05	6.70e+05	3.26e+05	3.53e+05	1.22e+05
F142017	2017	50	1.69e+09	6.59e+09	3.57e+09	3.68e+09	9.20e+08	5.73e+08	1.41e+09	1.02e+09	9.97e+08	2.09e+08	1.75e+05	4.71e+05	2.64e+05	2.77e+05	5.99e+04
F152017	2017	50	1.45e+08	1.83e+09	8.09e+08	8.33e+08	3.22e+08	1.64e+07	2.32e+08	1.11e+08	1.17e+08	4.30e+07	4.58e+03	8.90e+04	1.44e+04	2.39e+04	2.23e+04
F162017	2017	50	4.73e+09	3.84e+11	8.03e+10	9.91e+10	8.06e+10	1.08e+07	4.14e+09	6.27e+08	1.06e+09	1.06e+09	3.75e+02	3.25e+03	1.51e+03	1.58e+03	6.61e+02
F172017	2017	50	2.46e+07	1.32e+08	7.29e+07	7.34e+07	2.62e+07	7.37e+06	4.54e+07	1.83e+07	1.90e+07	6.46e+06	2.77e+05	1.16e+06	5.79e+05	6.14e+05	1.85e+05
F182017	2017	50	3.41e+11	4.12e+13	5.15e+12	8.13e+12	8.70e+12	8.38e+10	4.48e+11	2.08e+11	2.15e+11	8.48e+10	1.28e+05	3.80e+05	2.48e+05	2.44e+05	6.21e+04
F192017	2017	50	4.46e+03	9.84e+03	7.29e+03	7.31e+03	1.21e+03	2.69e+03	4.99e+03	3.53e+03	3.54e+03	5.44e+02	6.98e+01	7.74e+02	3.29e+02	3.07e+02	1.60e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	6.33e+03	9.29e+04	6.57e+04	5.37e+04	2.88e+04	3.70e+03	3.72e+04	2.71e+04	2.15e+04	1.12e+04	1.43e+02	7.19e+02	5.74e+02	4.73e+02	2.05e+02
F212017	2017	50	7.26e+02	1.24e+03	9.62e+02	9.55e+02	9.92e+01	4.88e+02	5.98e+02	5.43e+02	5.46e+02	2.58e+01	1.08e+02	2.33e+02	2.01e+02	1.82e+02	4.40e+01
F222017	2017	50	3.88e+04	8.83e+04	4.77e+04	5.54e+04	1.51e+04	2.25e+04	3.64e+04	2.74e+04	2.76e+04	2.83e+03	4.03e+02	9.64e+02	4.88e+02	5.71e+02	1.62e+02
F232017	2017	50	1.98e+04	5.25e+04	2.52e+04	2.83e+04	8.99e+03	1.21e+04	2.29e+04	1.46e+04	1.54e+04	2.22e+03	3.18e+02	6.77e+02	3.64e+02	4.08e+02	1.11e+02
F242017	2017	50	5.99e+03	1.28e+04	7.80e+03	8.00e+03	1.47e+03	2.08e+03	4.24e+03	2.85e+03	2.89e+03	4.35e+02	7.04e+02	1.01e+03	7.36e+02	7.51e+02	4.86e+01
F252017	2017	50	3.09e+03	6.97e+03	3.91e+03	3.96e+03	6.67e+02	2.02e+03	2.43e+03	2.22e+03	2.22e+03	9.35e+01	1.11e+03	1.65e+03	1.38e+03	1.38e+03	1.88e+02
F262017	2017	50	1.03e+03	1.66e+03	1.31e+03	1.33e+03	1.36e+02	6.79e+02	9.64e+02	8.50e+02	8.45e+02	6.07e+01	4.86e+02	5.82e+02	5.23e+02	5.26e+02	2.02e+01
F272017	2017	50	1.27e+03	2.65e+03	1.64e+03	1.69e+03	2.69e+02	6.81e+02	9.31e+02	7.99e+02	8.07e+02	5.45e+01	1.90e+01	4.54e+02	2.50e+01	4.80e+01	9.45e+01
F282017	2017	50	3.70e+10	6.56e+12	6.20e+11	9.39e+11	1.07e+12	3.73e+09	8.78e+10	2.64e+10	3.06e+10	1.77e+10	1.56e+04	1.01e+05	4.04e+04	4.50e+04	2.07e+04
F292017	2017	50	2.66e+10	5.71e+12	1.21e+12	1.60e+12	1.34e+12	2.66e+10	1.13e+11	7.38e+10	7.09e+10	2.36e+10	6.74e+05	1.02e+07	4.20e+06	4.49e+06	2.41e+06
F12022	2022	20	1.22e+04	3.06e+04	1.89e+04	1.91e+04	4.15e+03	3.21e+03	7.08e+03	5.43e+03	5.37e+03	9.92e+02	3.02e-01	8.32e-01	5.09e-01	5.23e-01	1.12e-01
F22022	2022	20	2.45e+02	6.58e+02	4.47e+02	4.60e+02	8.89e+01	1.25e+02	2.33e+02	1.66e+02	1.68e+02	2.45e+01	4.49e+01	4.91e+01	4.49e+01	4.67e+01	2.09e+00
F32022	2022	20	2.35e-01	4.81e-01	3.44e-01	3.45e-01	5.61e-02	8.46e-02	1.70e-01	1.38e-01	1.39e-01	1.97e-02	8.97e-06	2.35e-05	1.65e-05	1.61e-05	3.35e-06
F42022	2022	20	1.99e+02	3.49e+02	2.91e+02	2.89e+02	3.20e+01	1.54e+02	2.47e+02	2.09e+02	2.09e+02	2.18e+01	3.80e+01	1.38e+02	7.40e+01	7.42e+01	2.36e+01
F52022	2022	20	3.76e+00	8.39e+00	6.11e+00	6.18e+00	9.46e-01	1.46e+00	3.36e+00	2.43e+00	2.42e+00	4.03e-01	2.96e-04	6.71e-04	4.41e-04	4.44e-04	8.31e-05
F62022	2022	20	1.42e+08	8.15e+08	3.75e+08	3.89e+08	1.36e+08	3.71e+07	1.79e+08	1.14e+08	1.10e+08	3.81e+07	3.27e+04	1.48e+05	8.70e+04	8.71e+04	2.33e+04
F72022	2022	20	3.99e+02	1.68e+03	1.07e+03	1.05e+03	2.46e+02	2.21e+02	6.67e+02	4.31e+02	4.45e+02	9.71e+01	2.92e+01	7.73e+01	3.32e+01	3.46e+01	6.87e+00
F82022	2022	20	9.84e+03	1.17e+08	6.19e+06	1.54e+07	2.25e+07	1.71e+03	4.57e+05	9.84e+03	3.33e+04	7.01e+04	2.54e+01	3.85e+01	2.76e+01	2.82e+01	2.27e+00
F92022	2022	20	4.42e+02	9.25e+02	5.84e+02	5.98e+02	7.64e+01	3.80e+02	4.58e+02	4.09e+02	4.11e+02	1.61e+01	3.36e+02	3.36e+02	3.36e+02	3.36e+02	1.79e-01
F102022	2022	20	4.89e+02	5.00e+03	5.35e+02	1.06e+03	1.23e+03	4.40e+02	4.44e+03	4.64e+02	8.24e+02	9.80e+02	3.95e+02	1.69e+03	4.00e+02	4.89e+02	2.30e+02
F112022	2022	20	8.25e+01	1.83e+03	1.45e+02	2.13e+02	3.28e+02	3.51e+01	6.85e+01	5.34e+01	5.33e+01	7.35e+00	1.42e-01	2.34e-01	1.80e-01	1.81e-01	1.93e-02
F122022	2022	20	2.92e+02	4.48e+02	3.35e+02	3.44e+02	3.35e+01	2.63e+02	3.27e+02	2.71e+02	2.74e+02	1.23e+01	2.35e+02	2.63e+02	2.39e+02	2.40e+02	3.81e+00